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Compound effects of thermal stress and tissue abrasion on photosynthesis and respiration in the reef-building coral *Acropora cervicornis* (Lamarck, 1816)

Bennett T. Paradis, Raymond P. Henry, Nanette E. Chadwick*

Department of Biological Sciences, 101 Rouse Life Sciences Building, Auburn University, Auburn, AL 36849, United States of America

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ABSTRACT

Reef-building corals are threatened by multiple stressors, including rising ocean temperatures due to anthropogenic climate change, and tissue abrasion inflicted by increasing frequencies of recreational diving on tropical reefs. Both can impair coral growth and survival, but interactive effects of temperature and abrasion remain unknown. We conducted laboratory experiments to investigate the metabolic physiology of endangered Caribbean staghorn corals *Acropora cervicornis* (Lamarck, 1816) in response to both types of stressors: (1) temperature variation (22, 25, 28, and 30 °C) and (2) tissue abrasion (15% of tissue removed vs. non-abraded controls). Using flow-through respirometry, we measured net photosynthesis and respiration in 26 coral fragments (6 genotypes) exposed to 8 treatment combinations (4 temperature × 2 abrasion treatments). In non-abraded corals, net photosynthesis peaked at 25 °C but respiration increased linearly with temperature. Higher temperatures caused a decline in the ratio of photosynthesis to respiration (P:R), wherein corals were unable to meet their metabolic energy needs (P:R $<$ 1). In addition, there was a significant interaction effect between the two stressors, in which tissue abrasion impaired photosynthesis and augmented respiration, especially at higher temperatures. We conclude that under conditions of both diver damage and elevated temperature, *A. cervicornis* cannot meet its energetic demands. These results indicate impairment of coral resilience to temperature-induced bleaching in areas exposed to high levels of abrasion (i.e.: popular dive sites). This information provides a scientific basis to support local reef management during climate change, and also reveals the metabolic processes that precede and underlie coral bleaching on tropical reefs.

1. Introduction

Coral reefs harbor complex symbiotic systems that support high levels of biodiversity in mostly nutrient poor environments; they are inhabited by members of 30 animal phyla, and are considered to be the ocean's equivalent of tropical rain forests (Birkeland, 2015). Tropical reefs are highly productive ecosystems and play a large role in carbon sequestration (Bellwood et al., 2004; Lamb et al., 2014). They additionally provide other benefits like storm protection to island and coastal communities, and are of socioeconomic importance to local populations (Bellwood et al., 2004; Lamb et al., 2014). However, a combination of rising ocean temperatures, acidification, terrestrial nutrient runoff, and physical damage from harmful fishing practices and human recreational activities have altered the state of tropical reefs in recent decades and are expected to inflict further damage in the near future (Guilherme Becker et al., 2013; Lamb et al., 2014; Orr et al., 2005; Raymundo et al., 2009). Their ongoing loss can be attributed partly to a disruption of the physiological mechanisms of calcareous reef-builders, which support the processes underlying the productive symbiosis between the coral host and microalgae. Thermal stress can result in a dramatic loss of endosymbionts, known as coral bleaching, and has become increasingly widespread. As ocean temperatures continue to rise above summer norms, global scale epidemics of bleaching on reefs have occurred in 1998, 2010, and 2015 (Hughes et al., 2017). During a bleaching event, oxidative stress and the buildup of free radicals cause damage to photosystem II (PSII) and its repair system in the coral's microalgae, resulting in expulsion of microalgal symbionts from the coral animal tissue (Brown et al., 2002; Warner et al., 1999; reviewed in Weis, 2008). This mechanism suggests that changes in physiological processes including photosynthesis and respiration precede the bleaching event, and reach a critical point which may trigger the expulsion of symbionts (Weis, 2008).

* Corresponding author.
E-mail address: chadwick@auburn.edu (N.E. Chadwick).

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Recent studies using respirometry to measure coral photosynthesi and respiration have documented changes in coral physiological processes under elevated temperature and irradiance (Al-Sofyani and Floos, 2013; Hoogenboom et al., 2006; Schutter et al., 2008; Ulstrup et al., 2011). While these types of measurements provide information about the energy input and output of corals under conditions of climate change, little is known about other compounding environmental factors that may interact with temperature stress. One of those factors is global ecotourism, which has generated a large recreational diving industry that places divers in close contact with coral reefs (Krieger and Chadwick, 2013; Renfro and Chadwick, 2017). At intensively-visited reef sites, divers inadvertently cause physical damage in the form of both sediment resuspension that smothers corals, and abrasion that removes coral tissues (Guzner et al., 2010; Lamb et al., 2014; Zakai and Chadwick-Furman, 2002). Synergistic effects of temperature extremes in combination with tissue abrasion on reef corals are expected to become increasingly common, especially on shallow reefs exposed to both intensive ecotourism and warming summer temperatures, but their interactive impact on coral physiology remain unexplored.

Staghorn coral (*Acropora cervicornis*) historically was one of the major shallow-water reef-builders in the Caribbean Sea. *Acropora cervicornis* and congener *A. palmata* (elkhorn corals) are the only 2 stony coral species that form large branching colonies in the Caribbean, and both enhance habitat complexity, biodiversity, and fish abundance on reefs (Dixson and Hay, 2012; Holbrook et al., 2002; Johnson et al., 2011). Due to population declines, both *A. cervicornis* and *A. palmata* are classified as critically endangered under the IUCN Red List. These and other species in the coral genus *Acropora* are highly susceptible to disease, temperature-induced bleaching, and other damaging agents (Aronson and Precht, 2001; Birkeland, 2015; Muller et al., 2008; Rogers, 1990), and due to their delicate structure remain the most-frequently damaged corals by recreational divers (Riegl and Velmirov, 1991; Rouphael and Inglis, 2002). While several studies have measured photosynthesis and respiration in stony corals, few have examined these processes in *A. cervicornis* (Davies, 1980), and none have determined physiological impacts of tissue abrasion in acroporid corals.

Here we describe interactive impacts of tissue abrasion and temperature extremes on photosynthesis and respiration in endangered staghorn corals *A. cervicornis*. We hypothesize that tissue abrasion and the resulting metabolic demands of wound healing act synergistically with thermal stress to hinder the ability of these corals to meet their metabolic energy demands, rendering them more susceptible to bleaching and mortality at high temperatures. Our results indicate that physiological damage to corals from global seawater temperature increases alone are enough to put them in a negative energy state, but these effects are further compounded by local disturbances in the form of tissue abrasion. Given that *Acropora* corals are dominant frame-builders on many shallow reefs (Lamb et al., 2014; Sheppard et al., 2009), and that diver-induced abrasion stress can become chronic on heavily-dived reefs (Gil et al., 2015; Hawkins et al., 1999; Lamb et al., 2014; Rouphael and Inglis, 2002), the processes revealed here suggest that coral reefs exposed to intensive diving tourism may be less resilient to global climate change.

2. Materials and methods

2.1. Coral collection and culture

Ten large fragments (each 18–28 cm length) of endangered scleractinian corals *Acropora cervicornis* (Lamarck, 1816) were obtained from the Smithsonian Marine Station (SMS), Fort Pierce, Florida, in February 2017. The fragments originated from two genetically identical colonies (genotype S1) collected in 2002 at ~10 m depth and cultured in the SMS public aquarium (25–26 °C aquarium seawater temperature during the month of shipping; William Hoffman, SMS, pers. comm.) before being shipped to Auburn University (AU) overnight in bags filled with seawater.

To provide a wider genetic diversity of coral fragments for experimentation, in March 2018, 25 small fragments (each 5–10 cm length) were shipped to AU overnight wrapped in damp paper towels (dry shipping method) from colonies cultured in offshore nurseries at 6–10 m depth (25–26 °C ambient seawater temperature at nurseries during the month of shipping) by the Coral Restoration Foundation (CRF), Key Largo, Florida, USA. Dry shipping was used for the latter shipment because it enhances coral fragment survival and is standard shipping procedure by CRF (Kayla Ripple, CRF, pers. comm.). This shipment included 5 fragments each of 5 genetically-distinct genotypes (25 fragments total), based on molecular analysis (genotypes B8, K1, M10, M1, and M6; Amelia Moura, CRF, pers. comm.).

Upon arrival at AU, coral fragments were acclimated over ~1 h to the conditions in laboratory culture tanks (8–12 fragments per tank × 4 tanks; genotypes interspersed randomly among tanks). All randomization of fragments, including for exposure to experimental treatments, was performed using a random number generator. Each 75 L culture tank was connected to a 75 L sump that provided filtration via protein skimmers (Class 110 INT Skimmer, Reef Octopus) and bioballs (Coral Life), and was maintained at 25 ± 1 °C and 34 ppt salinity (after Cantrell et al., 2015; Huebner et al., 2012; Roopin and Chadwick, 2009; see also DeSalvo et al., 2010). We considered 25 °C as the control temperature for the present study, because this was the seawater temperature at the sites of collection as well as in the culture tanks, and is known as an optimal temperature for tropical stony coral growth (Hubbard, 2015; Lirman et al., 2011; Seveso et al., 2016). Salinity was adjusted using Instant Ocean Sea Salt Mix (Instant Ocean), with a 30 L (20%) water change in each culture tank every 2 wk. Calcium, alkalinity, and magnesium were maintained at coral reef levels using liquid additives (calcium and alkalinity: Reef Fusion, Seachem; magnesium: Tech M Magnesium, Kent Marine). To enhance coral growth and survival, the water flow level was augmented by attaching a small pump to the inside wall of each culture tank, more closely simulating natural conditions in the exposed reef habitats where these corals naturally occur (Borneman, 2001; Jokiel, 1978; Roberts et al., 1992; Sheppard et al., 2009).

The corals were acclimated over 1 wk to irradiance levels in the culture tanks, because stony corals require several days to acclimate to changes in irradiance (Anthony and Hoegh-Guldberg, 2003; Titlyanov et al., 2000). Photosynthetically active radiation (PAR; 12 h:12 h light:dark cycle) was provided by fluorescent T5 light fixtures (Model: 3′ 6-bulb, 120 V, 960020, Sunlight Supply) installed over each tank, and was measured at the upper surface of each coral fragment using a Quantum Scalar Laboratory sensor (QSL-2101, Biospherical Instruments Inc.). Corals initially were exposed to low PAR (80 ± 20 mmol photons $m^{-2} s^{-1}$, mean + SD), gradually increasing over several days to moderate PAR (180 ± 20 mmol photons $m^{-2} s^{-1}$), equivalent to irradiance averaged over the entire day at ~18–23 m depth on Florida coral reefs (Lesser, 2000). They were fed weekly with a mixture of Coral Frenzy (Marine Depot) and live *Artemia* brine shrimp hatched from dried eggs (San Francisco Bay Brand). Higher irradiance levels were not used due to negative impacts of high irradiance (including UV) on corals under culture conditions (Baird et al., 2008; Lesser, 1997; Lesser, 2000; Levy et al., 2004).

Two weeks after arrival and acclimation to laboratory conditions, the large SMS fragments were divided using industrial pruning shears (Model #BP 7100D, Corona) into smaller fragments (S1a-k, each 6–15 cm length; see supplementary file, Tables S1 and S2) to allow them to fit into respirometry chambers (see below). The base of each fragment was attached to a circular ceramic plug (3.12 cm diameter × 1.3 cm height, Marine Depot) using cyanoacrylate gel (Reef Glue, Seachem). In each culture tank, a resting platform to hold the corals was created from white plastic egg crate grating (1.5 cm × 1.5 cm interior squares, Plaskolite) and PVC pipes. Coral fragments attached to their ceramic bases were placed in squares of this

grating, with the platform ~8 cm above the bottom of each culture tank.

Most cultured corals expanded their polyps and exhibited growth in buoyant weight, indicating healthy physiological condition (details in Paradis, 2018). After ~30–60 d, both their soft tissues and calcareous skeletons began to grow over the epoxy and ceramic plug at the base of each fragment. Coral condition also was gauged by color, recorded as the closest hue to that on a standard 6-hue color scale used for approximating the health of reef-building corals (CoralWatch, University of Queensland, coralwatch.org).

2.2. Coral size parameters

Four types of size parameters were calculated for each coral fragment: buoyant weight (BW), total length (TL), estimated surface area (ESA), and precise surface area (PSA). BW was determined monthly to the nearest 0.1 g for each fragment to assess growth as an indicator of physiological condition during 4–12 mo of laboratory culture prior to experimentation (after Jokiel, 1978). Immediately prior to the use of each fragment in a respirometry experiment (details below), TL was determined to the nearest mm using a flexible measuring tape to allow the tape to curve along the length of the fragment. ESA was calculated as previously described for *Acropora* spp. corals (Dennison and Barnes, 1988), and provided similar surface area estimates to those obtained using advanced 3D scanning techniques of the coral surface (Naumann et al., 2009). Given the limited number of specimens available due to the critically endangered status of *A. cervicornis* (IUCN Red List), we used ESA as a proxy for PSA during initial estimates of photosynthesis and respiration. This allowed us to obtain repeated respirometry measurements on each fragment before they were sacrificed for determination of PSA at the end of the study. To calculate PSA, the live tissue was removed from each fragment using a Water Pik (Waterpik Water Flosser, Ultra, Rite Aid; after Johannes and Wiebe, 1970; Richier et al., 2003), and the tin foil method was employed (after Marsh, 1970). Three metrics of body size were used to calculate the precise surface area of each fragment at the time of each respirometry experiment (PSA_t): buoyant weight (BW_t) at time of experiment, final buoyant weight at end of study (BW_f), and final precise surface area (PSA_f); assuming a proportional relationship between buoyant weight and surface area throughout a fragment's time in the culture tanks (Eq. (1)):

$$\frac{PSA_t\ (cm^2)}{BW_t\ (g)} = \frac{PSA_f\ (cm^2)}{BWT_f\ (g)} \quad (1)$$

2.3. Abrasion and temperature treatments

Coral fragments that were of appropriate size and shape to fit in the respirometry chamber (see below), and which grew over the ceramic base, exhibited consistent growth in terms of BW, and had a color hue of at least D3 or C3 on the color scale (CoralWatch, University of Queensland, coralwatch.org) were considered to be in good condition, and were used in experiments. Of the 26 small SMS fragments (created from 10 original large fragments) and 23 small CRF fragments (from 25 original small fragments) that survived initial acclimation to the laboratory, 35 small coral fragments (each 5–13 cm length; 12 SMS plus 23 CRF) remained in culture after 4–12 mo. Based on the above criteria, 26 of these remaining fragments (11 SMS and 15 CRF) were used in experiments, with at least 10 fragments representing 5–6 genotypes used in each of the 8 treatment combinations (see supplementary file, Tables S1 and S2). A power analysis was performed to determine the minimum required sample size for each treatment combination based on preliminary data, and was found to be $n > 8$ for power = 0.95 (https://clincalc.com/stats/samplesize.aspx).

To simulate the abrasion stress inflicted on corals when recreational divers and snorkelers scrape coral surfaces with their fins, hands, or other equipment (Krieger and Chadwick, 2013; Zakai and Chadwick-Furman, 2002), we applied an abrasion treatment. We initially manipulated 4 randomly-selected fragments to obtain preliminary information on how coral healing from abrasion wounds, and coral physiological responses, varied with wound size, and thus which wound dimensions to use for treatments. Two types of preliminary abrasion wounds were created: small (~1.5 cm^2 surface area) and large (~5 cm^2 surface area; n = 2 fragments each; each abraded area ~1.0 × 1.3 cm and 1.4 × 3.5 cm respectively), by scraping the coral surface with the blunt handle of a pair of metal forceps. Coral tissue was completely removed until only the underlying white calcium carbonate skeleton was visible in the abraded area, along the midsection of the long axis of each fragment. Photographs of abraded fragments were taken daily to gauge recovery time; fragments were considered to have recovered fully when they regained normal pigmentation and polyps were visible throughout the previously damaged area (see supplementary file, Fig. S1). Preliminary metabolic trials (see details below) were run to assess whether abraded corals altered their photosynthesis or respiration. Based on the results of these preliminary trials, fragments were selected randomly for assignment to 2 abrasion treatments: (1) Abraded (tissue removed from ~15–20% of fragment surface area), and (2) Non-abraded (no tissue removed; for sample sizes see also supplementary file, Tables S1 and S2). Abrasion wounds were created over ~15–20% of the surface area of each fragment (= ~2.5–8.4 cm^2 area, or wound dimensions of ~1.0 × 2.5 to 2.0 × 4.2 cm).

Abrasion treatments were initiated inside the culture tanks, then fragments were allowed a brief recovery period (72 h) before they were used in respirometry trials (details below). This recovery period allowed any loose damaged tissue or excess mucus along the margins of the abraded area to disperse and be filtered out of seawater in the culture tank, but not enough time for major healing of the abrasion wound prior to respirometry. Each fragment was abraded only once.

The dimensions of the abraded tissue area were measured to the nearest mm using calipers, and the abraded surface area (ASA) was calculated (= length × width of the wound). The ASA then was subtracted from ESA and PSA of the entire fragment when calculating photosynthesis and respiration, so they were reported per unit of live coral tissue. As such, calculations were corrected for the removed tissue, and measured oxygen fluxes only of the coral surface that remained covered with living tissue.

Fragments also were assigned randomly to 4 temperature treatments (22, 25, 28, and 30 °C) during respirometry trials, to test for interaction effects between these 2 types of stressors (8 treatment combinations; see supplementary file, Tables S1 and S2). Temperature treatments were selected based on the temperature range recorded on Florida coral reefs over the past 10 yr (US DOC/NOAA/NWS/National Data Buoy Center). The bleaching threshold for corals typically occurs at 1–2 °C above summer maximum temperatures (Baker et al., 2008; Hughes et al., 2003; Jokiel, 2004). Because the culture tanks were maintained at 25–26 °C, and summer sea surface temperature maxima in the Florida Keys are similar (US DOC/NOAA/NWS/National Data Buoy Center), 28 °C was considered to be a threshold level for bleaching, and fragments were expected to show metabolic responses indicative of a bleaching event. Recent coral bleaching events in combination with IPCC models also suggest that temperatures may rise beyond the 1–2 °C bleaching threshold, hence 30 °C was added to the temperature trials (Baker et al., 2008; Hughes et al., 2017; IPCC, 2007). On the other end of the spectrum, we chose 22 °C to reflect common minimum temperatures in the Florida Keys during winter months (US DOC/NOAA/NWS/National Data Buoy Center).

Due to the limited number of fragments available based on the endangered status of *A. cervicornis*, a repeated measures design was used for respirometry measurements. Fragments were selected randomly for treatment combinations (see Supplementary file, Tables S1 and S2), and potential confounding effects of any previous trials on the same fragment were accounted for in the statistical models (see

statistical analyses below). After a fragment had been randomly selected for abrasion (whether or not it had been selected previously as a non-abraded control coral, see Supplementary file, Table S2), it subsequently was removed from the selection pool for the non-abraded (control) treatment, to prevent any post-wound stress from affecting the responses of control (non-abraded) corals. Abraded fragments that were used in multiple temperature treatments underwent all trials within 3–4 d after wound infliction, and prior to any visible wound healing (see Supplementary file, Fig. S1). Because the SMS fragments (genotype S1) were obtained before the CRF fragments, many of the S1 fragments (S1a–S1f) were exposed to all 8 possible treatment combinations, in contrast to the CRF fragments which were exposed to a more limited set of treatments (see Supplementary file, Table S2).

2.4. Respirometry measurements

Metabolic responses of coral fragments assigned to the above abrasion and temperature treatments were examined using flow-through respirometry trials conducted intermittently during November 2017 to July 2018. A custom-made respirometry chamber was designed to hold each coral fragment in an upright position, and to accommodate the ceramic plug to which each fragment was attached (see Supplementary file, Fig. S2). Corals were starved for ~48 h before each respirometry trial to ensure that they exhibited basal metabolism. Coral fragments were transferred individually from their culture tanks into the respirometry chamber. Before placing each fragment into the chamber, the following variables were recorded while the fragment remained submerged in its culture tank: color (using a standard 5-hue color scale, see Section 2.1), total length (TL), and maximum diameter (each to the nearest 1 mm using flexible measuring tape. These numbers, along with the most recent monthly buoyant weight (BW), were used to calculate ESA and PSA. Any macroalgae or other organisms attached to the ceramic plug were scrubbed off using a stiff toothbrush, to prevent non-coral organisms from affecting metabolic measurements. Then while still submerged in the culture tank, the ceramic plug was fitted onto the Parafilm-covered rubber stopper from the base of the respirometry chamber, and the chamber tube and top stopper were placed over the entire fragment and plug, trapping some of the tank seawater inside the chamber. This process kept the coral fragment completely submerged in seawater and prevented any shocks due to rapid change in seawater temperature or exposure to air.

With the coral fragment inside, the chamber then was attached to the respirometry setup. Water flow was set at 30 mL min^{-1} to gradually flush the retained seawater from the culture tank out of the chamber (~15 min for 440 ml chamber volume). During initial water flushing, the coral fragment also was acclimated to the seawater temperature for its assigned treatment. A rapid change in water temperature was employed (up to 5 °C change in ~40 min, from 25 to 30 °C; after Coles and Jokiel, 1977). While this did not mimic climate change in time scale, it reflected the rapid temperature change that corals may experience naturally on reefs in enclosed areas such as lagoons (McCabe et al., 2010; Ouillon et al., 2005), a short-term process that is likely to become augmented as reefs experience ever more frequent and extreme high temperature events seasonally during global climate change (Hughes et al., 2003; Oliver and Palumbi, 2011). The water in the chamber was checked every 15 min during temperature acclimation, by opening the top of the chamber and inserting a digital thermometer (Model 9840, Taylor).

The water flow then was adjusted to ~8–13 mL min^{-1} for the duration of ~ 2.5 h incubation to measure net photosynthesis (after Jokiel and Coles, 1977; Hoogenboom et al., 2006; Schutter et al., 2008). An adjustable LED light fixture (Model HYA05-LENS-55*3W-B, Galaxyhydro) was set up lateral to the chamber, to evenly expose the long axis of the coral fragment to irradiance. Irradiance level was measured inside the chamber by filling it with seawater prior to trials, and inserting the probe of a Quantum Scalar Laboratory sensor (QSL-2101, Biospherical Instruments Inc.). The light then was adjusted so that 350 ± 20 mmol photons $m^{-2} s^{-1}$ reached the chamber interior (approximate irradiance during the midday maximum on Florida coral reefs at ~18–23 m depth; Lesser, 2000). This irradiance level was ~2× that in the culture tanks (see Section 2.1), so that the corals were exposed to irradiance conditions during midday on the coral reef. This method was expected to cause the corals to photosynthesize more rapidly than when in their culture tanks, i.e., at mid-day levels during their short period inside the metabolic chamber. The elevated irradiance was not expected to cause stress, because it was within the daily range of variation (10–1243 mmol photons $m^{-2} s^{-1}$ at 3 m depth, to 45–446 mmol photons $m^{-2} s^{-1}$ at 18 m; Lesser, 2000) on reefs where these fragments were acclimated prior to collection (see details above).

Before each trial, both oxygen electrodes were calibrated with air-saturated seawater and water deoxygenated with sodium sulfite at the same temperature as in the chamber. After the coral fragment had acclimated for 2 h inside the chamber, with flowing seawater at the desired temperature, both the inflow and outflow oxygen levels were recorded every min for 30 min. Oxygen solubility values for each temperature were taken from Dejours (1975), to convert the observed ΔPO_2 to ΔO_2 concentration (μmol L^{-1}). Net photosynthesis was calculated using the average values of inflow and outflow O_2 observed during the 30 min incubation, water flow, and surface area of the coral fragment in order to obtain measures in umol O_2 $cm^{-2} h^{-1}$, similar to those reported in previous respirometry experiments on corals (Eq. (2), after Anthony and Hoegh-Guldberg, 2003; Lesser, 2000; Table 1; see supplementary file, Table S3):

$$\frac{\text{Average } \Delta O_2\ (\mu\text{mol L}^{-1}) * \text{Flow rate (mL min}^{-1}) * 60\ \text{min/hr}}{\text{conversion factor } 1000\left(\frac{\text{mL}}{\text{L}}\right) * \text{Fragment Surface Area (cm}^2)} \quad (2)$$

Coral respiration was measured during the daytime by covering the respirometry chamber with blackout cloth to simulate dark conditions. This process yielded accurate coral nighttime respiration, because preliminary trials revealed that dark respiration of *A. cervicornis* as measured during both day and night was similar, so these corals did not appear to exhibit any clear circadian rhythm of respiration. Previous studies also have revealed no significant difference between dark respiration measured during the daytime vs. nighttime for stony coral fragments (Muthiga and Szmant, 1987).

A respiration trial was run immediately after the photosynthesis trial for each fragment, while it was still in the chamber. The light fixture adjacent to the chamber was turned off and blackout cloth was wrapped completely around the chamber (after Coles and Jokiel, 1977; Szczebak et al., 2013). The coral then was allowed to acclimate to dark conditions for 2 h, by which time the oxygen levels appeared to restabilize. Then the oxygen inflow and outflow measurements again were recorded every min for 30 min, similar to the photosynthesis trials, and respiration was calculated using the same equation Eq. (2).

At the end of each paired photosynthesis and respiration trial, seawater temperature inside the chamber was gradually (in ~15–20 min) returned to the culture tank temperature. The entire chamber with the fragment remaining inside then was re-submerged in the culture tank, and left there for 15–20 min as the seawater within the chamber adjusted to the ambient tank temperature, to minimize any temperature shocks to the corals after respirometry trials. With the respirometry chamber inside the culture tank, the coral was removed underwater. To prevent accumulation of microorganisms and biofilm that might affect future respirometry measurements, the plexiglass chamber was washed with DI water and scrubbed thoroughly before the next trial. Effects of water-borne bacteria on respiration and photosynthesis measurements were considered to be negligible, because blank trials run in the chambers filled with seawater but lacking coral fragments showed almost no O_2 flux between inflow and outflow electrodes.

Table 1
Published literature on variation in metabolic responses of scleractinian corals to environmental parameters. Shown are 3 types of metabolic responses (net photosynthesis, respiration, and ratio of net photosynthesis to respiration [P:R]) and 2 types of environmental parameters (photosynthetically active radiation [PAR] and seawater temperature). Only values for healthy, undamaged corals are reported (i.e. those that were not subjected to special treatments such as abrasion). SD or SE = values reported using standard deviation (SD) or standard error (SE) around the mean. Temp = seawater temperature. n = sample size in terms of number of corals examined. "-" = not given.

Coral species	Net photosynthesis (μmol O_2 L^{-1} cm^{-2} h^{-1})	Respiration (μmol O_2 L^{-1} cm^{-2} h^{-1})	P:R	SD or SE	Temp (°C)	PAR (μmol photons m^{-2} s^{-1})	n	Source
Acropora cervicornis	0.67 (± 0.11)	0.33 (± 0.11)	2.54 (± 0.39)	SE	22	350 ± 20	10–15	Present study
Acropora cervicornis	1.14 (± 0.08)	0.57 (± 0.08)	2.08 (± 0.26)	SE	25	350 ± 20	10–19	Present study
Acropora cervicornis	0.47 (± 0.11)	0.72 (± 0.11)	0.70 (± 0.37)	SE	28	350 ± 20	10–18	Present study
Acropora cervicornis	0.21 (± 0.12)	0.81 (± 0.12)	0.30 (± 0.41)	SE	30	350 ± 20	10–13	Present study
Acropora cervicornis	–	0.35 (± 0.01)**	N/A	SD	28	0	2	Davies (1980)
Acropora palmata	–	0.84 (± 0.10)**	N/A	SD	28	0	2	Davies (1980)
Agaricia grahamae	–	0.14 (± 0.03)**	N/A	SD	28	0	3	Davies (1980)
Agaricia lamarcki	–	0.40 (± 0.08)**	N/A	SD	28	0	2	Davies (1980)
Agaricia tenuifolia	4.28 (± 0.17)	0.44 (± 0.15)	9.79[b]†	SD	28–32	max 1900	4	Lesser (1997)
Agaricia tenuifolia	2.73 (± 0.54)	1.11 (± 0.67)	2.46[b]	SD	28–35	Max 1900	4	Lesser (1997)
Agaricia undata	–	0.25**	N/A	SD	28	0	1	Davies (1980)
Fungia scutaria	–	–	3.89 (± 0.36)	SE	30	Midday in Hawaii	3–4	Coles and Jokiel (1977)
Galaxea fascicularis	0.54*	–	N/A		25.5–26.5	38	9	Schutter et al. (2008)
Galaxea fascicularis	2.46 (± 0.24)††	1.24 (± 0.18)	1.98[b]	SE	26	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Galaxea fascicularis	2.43 (± 0.28)††	1.1 (± 0.16)	2.21[b]	SE	31	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Galaxea fascicularis	4.62*	–	N/A		25.5–26.5	410	9	Schutter et al. (2008)
Montastraea annularis	–	0.46 (± 0.06) – 0.81 (± 0.32)**	N/A	SD	28	0	13	Davies (1980)
Montastraea faveolata	1.48 (± 0.12)	0.43 (± 0.12)	3.44[b]	SD	29–30	242[a]	6	Lesser (2000)
Montastraea faveolata	1.88 (± 0.14)	0.08 (± 0.14)	23.50[b]	SD	29–30	170[a]	6	Lesser (2000)
Montastraea faveolata	1.93 (± 0.07)	0.11 (± 0.06)	17.55[b]	SD	29–30	60[a]	6	Lesser (2000)
Montastraea faveolata	2.12 (± 0.14)	0.02 (± 0.13)	106.00[b]	SD	29–30	201[a]	6	Lesser (2000)
Montastraea faveolata	3.10 (± 0.17)	0.06 (± 0.16)	51.67[b]	SD	29–30	682[a]	6	Lesser (2000)
Montastraea cavernosa	–	0.21 (± 0.03) – 0.41 (± 0.08)**	N/A	SD	28	0	8	Davies (1980)
Montipora verrucosa	–	–	2.85 (± 0.17)	SE	30	Midday in Hawaii	3–4	Coles and Jokiel (1977)
Montipora verrucosa	–	1.44 (± 0.24)*	N/A	SD	26	0	4	Stambler et al. (1994)
Pocillopora damicornis	–	–	2.21 (± 0.19)	SE	30	Midday in Hawaii	3–4	Coles and Jokiel (1977)
Pocillopora damicornis	–	0.29 (± 0.14)**	N/A	SD	25	–	10	Al-Sofyani and Floos (2013)
Pocillopora damicornis	–	0.37 (± 0.10)**	N/A	SD	30	–	10	Al-Sofyani and Floos (2013)
Pocillopora damicornis	–	0.29 (± 0.10)**	N/A	SD	35	–	10	Al-Sofyani and Floos (2013)
Pocillopora damicornis	0.47 (± 0.07)***	1.01 (± 0.25)***	0.54 (± 0.16)	SE	24	900	4	Ulstrup et al. (2011)
Pocillopora damicornis	1.98 (± 0.90)***	1.30 (± 0.18)***	1.79 (± 0.99)	SE	28.5	900	4	Ulstrup et al. (2011)
Pocillopora verrucosa	–	0.31 (± 0.09)**	N/A	SD	25	–	10	Al-Sofyani and Floos (2013)
Pocillopora verrucosa	–	0.42 (± 0.10)**	N/A	SD	30	–	10	Al-Sofyani and Floos (2013)
Pocillopora verrucosa	–	0.41 (± 0.11)**	N/A	SD	35	–	10	Al-Sofyani and Floos (2013)
Stylophora pistillata	1.20 (± 0.18)††	0.64 (± 0.07)	1.88[b]	SE	26	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Stylophora pistillata	0.90 (± 0.11)††	0.70 (± 0.06)	1.29[b]	SE	31	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Turbinaria reniformis	3.23 (± 0.51)††	1.26 (± 0.17)	2.56[b]	SE	26	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Turbinaria reniformis	2.23 (± 0.30)††	1.10 (± 0.12)	2.03[b]	SE	31	300 (± 30)	8	Ferrier-Pagès et al. (2010)
Turbinaria reniformis	0.86 (± 0.29)***	0.79 (± 0.18)***	1.35 (± 0.53)	SE	24	900	4	Ulstrup et al. (2011)
Turbinaria reniformis	5.26 (± 2.23)***	3.28 (± 0.432)***	1.57 (± 0.69)	SE	28.5	900	4	Ulstrup et al. (2011)

* Originally reported as μmol O2 cm^{-2} min^{-1}.
** Originally reported as μL O2 cm^{-2} h^{-1}.
*** Originally reported as nmol O2 cm^{-2} s^{-1}.
[a] Calculated from averaged daily irradiance values.
[b] Calculated from given values.
† Originally reported as nmol O2 cm^{-2} h^{-1}.
†† Originally reported as Pgross (or Pmax), calculation made by the equation Pgross = Pnet + Rdark.

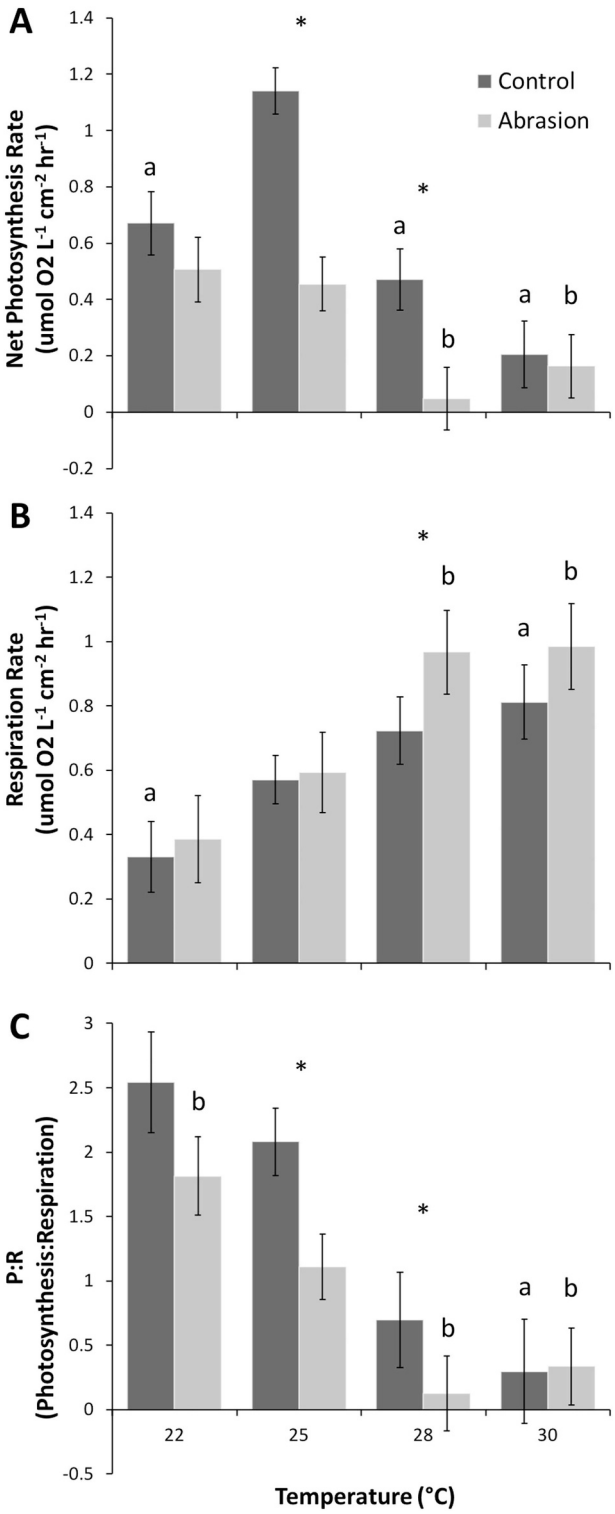

Fig. 1. Variation in metabolic responses of endangered staghorn coral *Acropora cervicornis* among fragments exposed to temperature and tissue abrasion treatments: (A) net photosynthesis, (B) respiration, and (C) P:R (ratio of photosynthesis to respiration, a measure of energy balance). 'a' and 'b' indicate responses that differed significantly from those observed at the control temperature (25 °C), for coral fragments that had not been abraded (n = 64 measurements on 26 fragments, with up to 4 measurements on some fragments) and for those that had received a tissue abrasion treatment (n = 40 measurements on 17 fragments, with up to 4 measurements on some fragments), respectively. *Indicates values that differed significantly between abraded and non-abraded coral fragments at each temperature; analyzed using linear mixed effects models with repeated ANOVAs. Shown are means ± 1 SE. See Section 2.5 for details on statistical tests applied, and Tables S1 and S2 for detailed sample

2.5. Statistical analyses

Relationships among coral size parameters were compared using Pearson correlation tests. To test for effects of the 4 temperature treatments on 3 metabolic parameters (net photosynthesis, respiration, and ratio of photosynthesis to respiration [P:R]), repeated measures ANOVAs were run using the linear mixed effects (lme) function within the nlme package of the statistical program R (R Core Team, 2016). This model included a random effect for individual fragment, to account for any variability contributed by a particular fragment being tested in multiple treatments (i.e., for a fragment used in trials at both 22 and 28 °C, any confounding effects of treatment order were accounted for in the model; see supplementary file, Tables S1 and S2). To determine the significance of interaction effects between temperature and abrasion treatments, a holistic linear mixed effects model was used initially to analyze all respirometry trials. Then because a significant interaction effect was detected between temperature and abrasion in terms of net photosynthesis (see Results), two linear mixed effects (lme) models were used to analyze the abrasion treatments separately. In these models, repeated measures ANOVAs tested for variation among metabolic parameters across the temperatures used, separately for the two abrasion treatments. In each model, 25 °C was used as the control, or reference, class.

To compare differences between non-abrasion and abrasion trials for the three metabolic parameters examined, four additional models were created, one for each experimental temperature. In these subsequent models, the data were included for both abraded and non-abraded fragments, and run for each of the 4 examined temperatures. To account for any variability contributed by fragments that were tested in multiple treatments, each model included a random effect for individual fragment. The data fulfilled assumptions of all the statistical models. Results are expressed as means ± 1 SE, unless otherwise indicated (similar to previous studies; Table 1).

3. Results

3.1. Coral size parameters

All 4 examined size parameters of fragments of staghorn corals *Acropora cervicornis* (buoyant weight [BW], total length [TL], estimated surface area [ESA], and precise surface area [PSA]) correlated significantly with each other ($r^2 > 0.90$ for most relationships, see supplementary file, Fig. S3). ESA was a better indicator of BW than was TL for these coral fragments, and as such was used to standardize measures of photosynthesis and respiration.

3.2. Coral growth and healing from tissue abrasion

Within 3–4 wk after arrival to the laboratory, the coral fragments produced new calcareous skeleton and soft tissue that extended over their ceramic bases. The polyps remained continually expanded on ~75% of the 26 small SMS fragments and 100% of the 23 small CRF fragments (49 fragments total) that survived initial laboratory acclimation and separation into small from large fragments. During 4–10 mo of laboratory culture (duration depending on whether they were from SMS or CRF, see Section 2.1), the coral fragments grew ~0.1–0.2 g mo^{-1}, as measured in BW. Fragments that grew relatively rapidly exhibited white or pale polyps at the fragment branch tip, whereas those that grew more slowly often exhibited terminal polyps that were darker in color.

Preliminary observations indicated that relatively large abrasion wounds (5 cm^2 surface area) fully recovered in ~45 d (range = 40–47 d, n = 2 fragments, see supplementary file, Fig. S3), while smaller abrasion wounds (1.5 cm^2) fully recovered in only ~32 d (31–39 d). As such, abrasion wound size for the experimental treatments was set at 15–20% of fragment total surface area (~2.5–8.4 cm^2

Table 2
Three types of metabolic parameters exhibited by fragments of endangered staghorn corals *Acropora cervicornis* exposed to a combination of temperature variation (4 treatments) and tissue abrasion (2 treatments), as determined by flow-through respirometry. Values are presented as mean and standard error (SE) for each treatment combination. n = number of respirometry trials per treatment combination. See text and Table S2 for details.

		Net photosynthesis (μmol O2 L^{-1} cm^{-2} h^{-1})		Respiration (μmol O2 L^{-1} cm^{-2} h^{-1})		P:R ratio		
Tissue abrasion treatment	Temperature treatment (°C)	Mean	SE	Mean	SE	Mean	SE	n
Non-abraded (control)	22	0.67	0.11	0.33	0.11	2.54	0.39	15
	25	1.14	0.08	0.57	0.08	2.08	0.26	19
	28	0.47	0.11	0.72	0.11	0.70	0.37	17
	30	0.21	0.12	0.81	0.12	0.30	0.41	13
Abraded	22	0.51	0.12	0.39	0.14	1.81	0.30	10
	25	0.46	0.10	0.59	0.13	1.11	0.25	10
	28	0.05	0.11	0.97	0.13	0.13	0.29	10
	30	0.16	0.11	0.98	0.13	0.34	0.30	10

surface area, or wound dimensions of ~1.0 × 2.5 to 2.0 × 4.2 cm).

3.3. Effect of temperature on non-abraded corals

There was a significant interaction effect between temperature and abrasion ($F_{3,72}$ = 4.683, p < .01) on net photosynthesis by the coral fragments, so abraded vs. non-abraded fragments were analyzed separately. There was no significant interaction effect between temperature and abrasion on either respiration ($F_{3,72}$ = 0.475, p = .70) or on ratios of photosynthesis to respiration (P:R, $F_{3,72}$ = 0.904, p = .44).

In non-abraded (control) corals (n = 64 trials on 26 fragments from 6 genotypes; see supplementary file, Tables S1 and S2), net photosynthesis peaked at 25 °C and decreased as temperature increased or decreased (Fig. 1A; Table 2), indicating that seawater temperature of 25 °C was the thermal optimum for net photosynthesis in these cultured fragments of *A. cervicornis*. Corals incubated at relatively low temperature (22 °C) exhibited a significant (on average 41%) decline in oxygen production vs. at the 25 °C thermal optimum (t = −4.136, p < .001), whereas those at relatively high temperature (28 °C) showed an even greater decline (59%) from the 25 °C optimum (t = −6.120, p < .001), and fragments at extremely high temperature (30 °C) corresponded to an 82% decrease from maximal values at the 25 °C thermal optimum (t = −7.902, p < .001). Net photosynthesis in corals at extremely high temperature (30 °C) also was significantly lower (by 69%) than that at relatively low temperature (22 °C, t = −3.805, p < .001).

In contrast, the respiration of non-abraded corals increased linearly with temperature (Fig. 1B) more gradually than did photosynthesis. At the relatively low temperature of 22 °C, O_2 consumption fell to significantly lower than that at 25 °C (by 42%, t = 2.187, p < .05), however at 28 °C it was only 27% greater than at 25 °C, and not significantly different (t = −1.443, p = .157). At extremely high temperature (30 °C), respiration reached its highest observed value, a significant increase of 42% over that at the 25 °C thermal optimum for photosynthesis (t = −2.092, p < .05).

Based on the above patterns, the ratio of photosynthesis to respiration (P:R) in non-abraded corals decreased significantly with temperature. Corals incubated at low temperature (22 °C) had P:R ratios approaching ~3:1, which decreased approximately linearly to P:R ratios < 1:1 at both of the high temperatures (28 and 30 °C; Fig. 1C). The highest P:R ratio was recorded at low temperature (22 °C; Fig. 1C). Although not significantly different, the mean P:R ratio was 22% lower at 25 °C than at 22 °C (t = 1.187, p = .243). Fragments at 28 °C further decreased their P:R ratio, dipping below the metabolic equilibrium of P:R = 1. As such, P:R at 28 °C was significantly less (by 66%) than at 25 °C (t = −3.727, p < .001). The lowest P:R ratio was exhibited by corals incubated at the highest temperature (30 °C). This corresponded to a significant decline of 86% from P:R at 25 °C (t = −4.388, p < .001).

3.4. Effect of temperature on abraded corals

In contrast to the pattern observed for non-abraded control corals, during the 40 respirometry trials (n = 10 trials at each of 4 temperatures) conducted on abraded corals (n = 17 fragments comprising 6 genotypes; see supplementary file, Tables S1 and S2), net photosynthesis peaked at 22 °C, then decreased gradually with temperature (Fig. 1A). When incubated at the control temperature of 25 °C, abraded coral fragments exhibited net photosynthesis which was not significantly higher than at cooler temperature (22 °C), only 11% difference in O_2 production, t = 0.447, p = .66). Net photosynthesis decreased sharply at elevated temperatures, to low values of almost no net oxygen production (28 °C: on average 90% lower than at 25 °C, t = −3.677, p < .005; 30 °C: on average 64% lower than at 25 °C, t = −2.576, p < .05; no significant difference between 28 °C and 30 °C, t = 0.971. p = .33; Table 2).

Similar to the pattern observed for non-abraded controls, the respiration of abraded coral fragments increased linearly with temperature (Fig. 1B). Abraded corals incubated at both 22 °C and 25 °C exhibited relatively low respiration (35% increase from 22 to 25 °C but not significantly different, t = 1.531, p = .14). When incubated at 28 °C and 30 °C, abraded corals significantly increased their respiration above that at 25 °C (63–66% higher than at 25 °C, t = −2.879 and − 2.931, respectively; p < .01 for both).

The ratio of photosynthesis to respiration (P:R) in abraded coral fragments decreased significantly with temperature, and was above energy compensation point (> 1:1) at only the 2 lowest temperatures examined (22 and 25 °C; Fig. 1C). P:R was much higher (63% on average) at 22 °C than 25 °C, but even so was barely above compensation point (significantly different, t = 4.374, p < .05). P:R decreased sharply at 28 °C, at which point the abraded corals exhibited negative net photosynthesis (i.e., net oxygen consumption) even under light conditions (significantly lower P:R than at 25 °C, t = −3.367, p < .005; representing a decline of 89% on average). P:R at the highest incubation temperature (30 °C) was not significantly different from that at 28 °C (t = −0.651, p = .52), and on average 70% less than at 25 °C (t = −2.600, p < .05).

3.5. Comparison of non-abraded and abraded corals

Comparison of metabolic performance in the non-abraded vs. abraded coral fragments revealed significant impairment of net photosynthesis at some treatment temperatures (Fig. 1A). Because the data for both photosynthesis and respiration were normalized to the surface area of living coral tissue (i.e., approximate area of the wound was subtracted from the total surface area on each abraded coral fragment, see Section 2.3) this difference indicates that wounds created by tissue abrasion reduced net oxygen production even in the non-damaged portions of the coral colony tissue. At 25 °C, abraded fragments cut their

net photosynthesis to less than half (60% less on average) that of non-abraded fragments ($t = -3.462$, $p < .05$), despite being incubated at the thermal optimum for photosynthesis. At 28 °C, abraded corals exhibited even greater impairment of net oxygen production (on average 91% less than that of non-abraded corals, $t = -3.829$, $p < .01$). Due to the relatively low photosynthesis exhibited by both abraded and non-abraded corals at more extreme temperatures (Fig. 1A), photosynthesis did not differ significantly at either 22 or 30 °C ($t = -1.403$ and -1.135, $p = .19$ and 0.30, respectively).

In terms of respiration, the corals that received abrasion treatments respired on average more rapidly per live tissue area than did non-abraded controls, at all 4 temperatures examined, and this difference was augmented by increasing temperature (Fig. 1B). However, the impacts of tissue abrasion on respiration were more subtle than on net photosynthesis, with significant elevation of respiration the in abraded corals only at 28 °C (by 35% over non-abraded corals, $t = -2.679$, $p < .05$). Multivariate modeling that accounted for random effects did not reveal significant differences at 22 °C ($t = 0.274$, $p = .79$), 25 °C ($t = -1.557$, $p = .16$), or 30 °C ($t = -0.814$, $p = .45$).

Because tissue abrasion caused significant depression of net photosynthesis and augmentation of respiration at moderate temperatures, the abraded fragments exhibited significantly lower P:R ratios than did non-abraded controls at both of those temperatures (25 °C: 47% lower; 28 °C, 82% lower; $t = -4.209$ and -4.689 respectively, $p < .005$ for both), but no significant difference at either of the more extreme temperatures (22 °C: $t = -0.424$, $p = .682$; 30 °C: $t = -1.724$, $p = .136$). Overall, both the temperature and tissue abrasion treatments significantly impacted all 3 metabolic parameters examined (photosynthesis, respiration, P:R ratio), with some interaction effects between the 2 types of stressors.

4. Discussion

We demonstrate here that peak photosynthesis in colonies of endangered staghorn coral *Acropora cervicornis* from the Florida Keys occurs at 25 °C, which is similar to median annual seawater temperature in the field, suggesting that these corals perform best at a temperature matching that to which they are field acclimated. Below and above this optimum, colonies experience declines in net photosynthesis, whereas respiration increases linearly with temperature. Our experimental manipulations reveal that a combination of 2 major stressors (high temperature and tissue abrasion) causes theses corals to enter an energy deficit, and that abrasion stress hinders their ability to produce enough energy for growth even at optimal temperature. These results have implications for the management of coastal reefs exposed to diving tourism or other local disturbances that abrade coral tissues, especially under conditions of high seawater temperature induced by anthropogenic climate change.

The tight correlation observed here between coral surface area estimates obtained using two major methods, the tin foil method (Marsh, 1970) for precise surface area (PSA) and the geometric equation method (Naumann et al., 2009) for estimated surface area (ESA), indicate that non-invasive geometric measurements can be used to estimate the size of some branching coral fragments with a high degree of accuracy. The tin foil method, similar to several other types of coral surface area measurement techniques, requires sacrificing the coral, whereas in corals with cylindrical branches, the dimensions of the branch's cylinder shell surface can be measured easily with minimal disturbance to the live coral. Naumann et al. (2009) confirmed that for branching corals in particular, the geometric equation provides surface area approximations that are similar to those obtained through advanced surface area determination techniques such as 3D scanning. Therefore, researchers who work with branch fragments of corals could effectively normalize their physiological and other data to unit surface area using this simple, non-invasive method to obtain ESA. Because the observed correlation between PSA and total length (TL) of the coral fragments was the second strongest correlation among the biometric variables examined, using TL as a proxy for PSA is the next best non-invasive method, especially in the field where branching corals typically have complex colony shape, and thus maximal branch diameter may be difficult to identify and measure.

The steady increase in coral fragment buoyant weight observed throughout the present study indicates that their physiological condition supported body growth during several months of laboratory culture, and that they were meeting their metabolic energy demands with enough surplus for growth. These corals thus were in robust health for our laboratory-based measurements of photosynthesis and respiration, and the levels obtained here were within the range of those reported in past studies on scleractinian corals under both laboratory and field conditions (Table 1).

Net photosynthesis of *A. cervicornis* was maximal at ~25 °C, similar to the results of other studies that have observed tropical coral photosynthesis to peak at moderate coral reef seawater temperatures of 25–27 °C, with declines at both cooler (22–24 °C) and warmer temperatures (~ 30 °C; Table 1; see also Saxby et al., 2003). Thus, the thermal optimum for net photosynthesis in this species (Coles and Brown, 2003; Howells et al., 2013) and for reef corals in general, appears to occur at moderate coral reef temperatures. This optimum varies somewhat among tropical regions, with deviations from the local optimum resulting in impaired coral primary productivity (Ferrier-Pagès et al., 2010; Lesser, 1997; Ulstrup et al., 2011). Tropical corals also grow most rapidly at ~26–27 °C in laboratory settings (Hubbard, 2015; Marshall and Clode, 2004; Precht and Aronson, 2004; Seveso et al., 2016). Coral specimens used in the present study thus appear to have been acclimatized to median annual field temperature observed on Florida reefs, and were maintained in the laboratory at their physiological temperature optimum.

We show here that in undamaged colonies of *A. cervicornis*, peak photosynthesis at moderate seawater temperature (~25 °C) combined with a linear increase in respiration with temperature leads to their ratio of photosynthesis to respiration (P:R) dropping below energy compensation threshold level at temperatures above ~28 °C. Our observation of impaired energetic output at extreme temperature parallels the known pattern of coral bleaching (expulsion of microalgae) at extreme sea surface temperature (SST; at both higher and lower than average field temperatures) on coral reefs worldwide. A 1–2 °C spike in summer average SST often leads to increased frequency of coral bleaching on reefs (Brown, 1997; Fitt et al., 2000; Hughes et al., 2017; Podesta and Glynn, 2001), with extremely cool temperature also causing bleaching in some regions (Hoegh-Guldberg et al., 2005; Lirman et al., 2011). These patterns suggest that impaired net photosynthesis of the microalgae harbored by corals is a major physiological mechanism underlying coral bleaching at extreme temperatures. Stress to the cellular processes involved in coral photosynthesis has been postulated as the reason for accumulation of reactive oxygen species (ROS) and expulsion of endosymbiotic microalgae that occur during bleaching (Chipuk and Green, 2008; Cunning and Baker, 2013; Dunn et al., 2007; Weis, 2008). At the cellular level, breakdown in the function of photosystem II (PSII) and its repair mechanisms at extreme temperatures (Douglas, 2003; Takahashi et al., 2004; Venn et al., 2008; Weis, 2008) likely contributed to the temperature-related patterns of coral photosynthesis observed here.

Respiration, on the other hand, increased linearly with seawater temperature, with the warmest examined conditions yielding the highest oxygen consumption, at values similar to those reported from previous studies on coral respiration (Table 1). This temperature-dependent pattern has been observed in all tropical stony corals examined to date: *Turbinaria reniformis*, *Pocillopora damicornis* (Al-Sofyani and Floos, 2013; Ulstrup et al., 2011), *Stylophora pistillata* (Ferrier-Pagès et al., 2010), *Siderastrea siderea* (Muthiga and Szmant, 1987), and *Agaricia tenuifolia* (Lesser, 1997). Augmented coral respiration at relatively high temperature is coupled with cellular stress responses, in

which corals initially up-regulate the production of heat shock proteins (HSPs) and antioxidants, both of which require energy consumption (Desalvo et al., 2010; Mayer, 2010; Rosic et al., 2011; Seveso et al., 2014). Therefore, it is likely that the increase in oxygen consumption at high temperature observed in the present study was caused in part by a thermally-induced stress response, wherein elevated temperature enhanced the energetically-expensive production of HSPs and antioxidants. The evolutionary history of organisms, and temperature ranges in the environments in which they evolved, both strongly influence their HSP expression, so studies on long-term physiological and evolutionary responses are needed to gain a realistic picture of how corals are likely to fare during climate change (Tomanek, 2010).

Our results reveal physiological processes exhibited by corals at high temperature prior to bleaching, which contribute to metabolic stress that is visually undetectable. Thus, compromised coral condition is a likely precursor to major loss of endosymbiotic algae in the form of colony bleaching. Because increased temperature puts the coral into negative energy balance (more energy consumption via respiration than production via photosynthesis; P:R < 1), at elevated temperature even if corals retain their microalgae (i.e.: have not reached a bleaching threshold), they have already entered a state of chronic malnutrition and are slowly starving. This observed state of reduced nutrition at sub-bleaching temperature renders corals more susceptible to additional stressors that could trigger the visibly obvious response of bleaching.

In addition to effects of temperature stress, our results revealed major impairment of net photosynthesis in corals exposed to tissue abrasion, in that abraded corals incubated near their temperature optimum (25 °C) exhibit levels of photosynthesis that are only ~1/3 (37%) those of non-abraded corals at the same temperature. These findings indicate that even when corals are not under thermal stress, tissue abrasion of only 15% of colony surface area significantly undermines the coral's ability to meet metabolic demands through photosynthesis. Additionally, abraded corals exposed to 28 °C exhibit net consumption of oxygen under daytime irradiance conditions, rather than the typical oxygen production response expected from photosynthetic organisms. Because the flow-through respirometer used in these experiments measured net photosynthesis (i.e.: gross photosynthesis – daytime respiration), this result suggests that daytime coral respiration swamps photosynthetic output in abraded corals at this temperature. Tissue abrasion also can encourage infection by coral disease, which would be another added stress to corals damaged by contact with recreational divers (Lamb et al., 2014).

As described by Coles and Jokiel (1977), P:R ratio is a dimensionless number that indicates the overall energy budget of the coral holobiont, regardless of biomass and skeletal size differences among corals. Examination of P:R ratio thus is useful for understanding how photosynthetic organisms meet their metabolic energy demands under stress conditions. The values obtained here for P:R of *A. cervicornis* fragments were consistent with those reported from previous studies on stony corals, although it is worth noting that a wide range of values has been reported in the literature (Table 1). For non-abraded coral fragments, the almost linear decrease observed in P:R with temperature indicates that at moderate temperatures the coral's energy needs are more than met through photosynthesis. The observed drop in P:R to values below 1 at warmer temperatures reveals that the coral's microalgae are producing less organic matter than required for cellular respiration of the holobiont (Coles and Jokiel, 1977). Overall, these patterns suggest that healthy, undamaged fragments of *A. cervicornis* are unable to meet their metabolic requirements when exposed to bleaching threshold temperatures on coral reefs (1–2 °C above mean summer SST; Cook et al., 1990; Fitt et al., 2000; Brown, 1997; Hughes et al., 2017). Previous studies have reported similar patterns, wherein P:R ratios were reduced at elevated temperatures in the stony corals *Agaricia tenuifolia* (Lesser, 1997), *Stylophora pistillata*, *Turbinaria reniformis*, *Galaxea fascicularis* (Ferrier-Pagès et al., 2010), *Pocillopora damicornis*, *Montipora verrucosa*, *Porites compressa*, and *Fungia scutaria* (Coles and Jokiel, 1977). We show here that a reduced P:R ratio at high temperatures in stony corals is caused by a significant decrease in photosynthetic activity coupled with normal temperature-dependent increases in aerobic metabolism. Because net photosynthesis declines far more rapidly than dark respiration increases with temperature, this pattern suggests that a breakdown in the photosynthetic pathway causes energy disequilibrium in corals exposed to thermal stress.

The corals subjected here to tissue abrasion showed a similar trend to that of non-abraded corals in terms of declining P:R ratio with temperature, but with more pronounced effects. At the moderate tropical temperature of 25 °C, the observed mean P:R value of barely ~1:1 suggests that even under optimal acclimated conditions, abraded corals generate barely enough energy to support general body maintenance and also meet additional energetic needs for tissue repair and skeletal regeneration. Of greater conservation concern is the observed pattern of abraded corals exhibiting net oxygen consumption at warmer temperatures (28 and 30 °C). This pattern suggests that if a colony becomes abraded prior to rise in seawater temperature, the combination of these 2 stressors could lead to a catastrophic sequence of events in terms of the coral's ability to meet its energy demands, in which it enters a state of negative energy balance with enhanced susceptibility to bleaching.

Both types of stressors applied here place additional energy demands on corals in terms of repairing tissue and re-growing their underlying calcium carbonate skeletons after abrasion, and also in terms of cellular responses to combat breakdown of the photosynthetic pathway and buildup of ROS following extreme temperature stress. Due to significant interaction effects between these two stressors on photosynthesis in *A. cervicornis*, it appears from an energetic standpoint that if corals are exposed to tissue abrasion, such as inflicted by recreational diver damage, they may become less resilient to climate change, in that they are more vulnerable to starvation in the event of a rise in sea temperature. Future research on both the short-term physiological responses of corals such as those examined here, in combination with their longer-term evolutionary responses, is expected to provide an increasingly realistic picture of how corals will continue to respond to both types of anthropogenic stressors.

Author contributions

B.T.P., R.P.H. and N.E.C. conceived the study and B.T.P. drafted the paper. B.T.P. performed the experiments, with the help of R.P.H. Data analysis was performed by B.T.P., with the help of R.P.H. and N.E.C. All authors participated in revising the manuscript and in obtaining the funding.

Data availability

All raw data have been deposited online as supplementary information at https://doi.org/10.1016/j.jembe.2019.151222.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jembe.2019.151222.

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